Customer Support Ticket Sentiment & Topic Analysis System

1. Introduction

In today's digital era, organizations receive a large volume of customer support tickets through various channels such as email, chat, and web forms. These tickets contain valuable information regarding customer issues, product defects, and service quality. However, manually analyzing these tickets is time-consuming, inconsistent, and inefficient.

The need for an automated system to process and analyze customer support tickets has become essential. Natural Language Processing (NLP) techniques enable machines to understand and interpret human language, making it possible to extract insights from textual data efficiently.

This project focuses on developing an intelligent system that automatically analyzes customer support tickets to identify sentiment and categorize them into appropriate departments. This helps organizations improve response time, prioritize critical issues, and enhance customer satisfaction.

2. Problem Statement

Many companies face challenges in handling large volumes of customer support tickets daily. The major issues include:

- Manual processing of tickets is slow and labour-intensive
- Inconsistent categorization due to human error
- Difficulty in identifying urgent or negative tickets quickly
- Lack of actionable insights from textual data

As a result, organizations experience delayed responses, missed critical complaints, and reduced customer satisfaction.

This project aims to solve these problems by developing an automated NLP-based system that can:

- Detect customer sentiment (positive or negative)
- Classify tickets into relevant departments
- Provide insights for better decision-making

3. Objectives

The main objectives of this project are:

- To build a system that automatically analyses customer support tickets
- To perform sentiment analysis to detect customer emotions
- To classify tickets into appropriate categories or departments
- To reduce manual effort and improve efficiency
- To provide a scalable and intelligent solution for businesses

4. Methodology

4.1 Data Collection

The dataset used in this project contains customer support tickets with the following attributes:

- Ticket Subject
- Ticket Description
- Customer Satisfaction Rating
- Ticket Type (Department)

The dataset is pre-processed and cleaned before applying machine learning techniques.

4.2 Data Preprocessing

Data preprocessing is an important step to prepare the text for model training. The following steps were performed:

- Conversion of text to lowercase
- Removal of special characters and symbols
- Combining subject and description into a single text field
- Handling missing values
- Tokenization using a pretrained tokenizer

Additionally, sentiment labels were generated based on customer satisfaction ratings:

- Rating ≥ 4 → Positive
- Rating ≤ 2 → Negative

Neutral values were removed to improve model clarity.

4.3 Model Design

A Transformer-based deep learning model was used for this project.

Model Used: DistilBERT

- A lightweight version of BERT (Bidirectional Encoder Representations from Transformers)
- Pretrained on large text corpora
- Provides high accuracy with reduced computation

Two models were implemented:

1. **Sentiment Classification Model**
    - Input: Ticket text
    - Output: Positive or Negative sentiment

2. **Topic Classification Model**
    - Input: Ticket text
    - Output: Department (e.g., Refund, Delivery, Product Inquiry)

4.4 Training Process

- Dataset split into training and testing sets (80:20 ratio)
- Tokenization using pretrained tokenizer
- Model fine-tuned on dataset
- Training performed using PyTorch in Google Colab
- Optimizer: AdamW
- Loss function: Cross-entropy

To speed up training, a subset of the dataset and reduced epochs were used.

4.5 Evaluation Metrics

The model performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

Additionally, visualization techniques were used:

- Confusion Matrix
- Heatmap

These metrics help in understanding model performance across different classes.

5. Results and Analysis

The system was successfully able to:

- Classify customer sentiment
- Assign tickets to appropriate departments

Performance:

- Fast model accuracy: ~75–85%
- Optimized model accuracy: up to ~90%

Observations:

- The model performs well for clear positive/negative sentences
- Some misclassifications occur due to:
    - Ambiguous text
    - Limited dataset size
    - Imbalanced data

Visualization:

- Confusion matrix shows correct and incorrect predictions
- Heatmap helps in identifying model weaknesses

6. System Implementation

The system is implemented in Google Colab using:

- Python
- PyTorch
- Hugging Face Transformers
- Scikit-learn
- Matplotlib & Seaborn

Workflow:

1. User inputs a support ticket
2. Text is cleaned and tokenized
3. Model predicts sentiment
4. Model assigns department
5. Output is displayed to user

7. Applications

This system can be used in:

- Customer support centers
- E-commerce platforms
- Service-based companies
- Helpdesk automation systems

8. Conclusion

The project successfully demonstrates the use of NLP and deep learning techniques to automate customer support ticket analysis. By using transformer-based models like DistilBERT, the system achieves high accuracy and efficiency.

The system helps organizations:

- Reduce manual workload
- Improve response time
- Identify critical issues quickly
- Enhance overall customer experience